

Distributionally Robust Optimization With Wasserstein Distance

We consider data-driven approaches to solving two optimization problems, a chance-constrained knapsack problem and a CVaR-constrained portfolio problem, based on distributionally robust formulations with Wasserstein distance, and investigate several validation schemes for selecting hyperparameters.

1 Optimization Problems

The chance-constrained (continuous) knapsack problem with d items is

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & P_F(\xi^T x \leq b) \geq 1 - \alpha \\ & 0 \leq x_l \leq 1 \text{ for each } l = 1, \dots, d \end{aligned} \tag{1}$$

where $c \in \mathbb{R}^d$ contains the losses of the items, $b \geq 0$ denotes the knapsack capacity, $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ is the decision, $\xi \in \mathbb{R}^d$ is a random vector that contains the item weights whose true joint distribution F is unknown, and $1 - \alpha$ is a given tolerance level (e.g., 90%). P_F denotes the probability under the true distribution F .

The CVaR-constrained portfolio problem with d assets is

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & \inf_{t \in \mathbb{R}} t + \frac{1}{\alpha} E_F[\max\{-\xi^T x - t, 0\}] \leq \gamma \\ & \sum_{l=1}^d x_l = 1, x_l \geq 0 \text{ for each } l = 1, \dots, d \end{aligned} \tag{2}$$

where $c = -E_F[\xi] \in \mathbb{R}^d$ (assumed known) now contains the expected losses of the assets, $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ is the allocation, and $\xi \in \mathbb{R}^d$ is the random vector of asset returns with unknown joint distribution F . The left hand side of the constraint represents the $(1 - \alpha)$ -CVaR of the portfolio loss, and γ is a given threshold for the CVaR. E_F denotes the expectation under the true distribution F .

2 DRO with Wasserstein Distance

2.1 Problem (1)

Let $\xi_1, \dots, \xi_n$ be a set of i.i.d. observations of ξ , and $\hat{F} = \frac{1}{n} \sum_{i=1}^n \delta_{\xi_i}$ be the empirical distribution, where δ_ξ denotes the Dirac measure at ξ . The distributionally robust optimization (DRO) formulation for (1) based on Wasserstein distance takes the form

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & \inf_{G \in \mathcal{B}(\hat{F}, \delta)} P_G(\xi^T x \leq b) \geq 1 - \alpha \\ & 0 \leq x_l \leq 1 \text{ for each } l = 1, \dots, d \end{aligned} \quad (3)$$

where $\mathcal{B}(\hat{F}, \delta)$ is the Wasserstein ball of radius $\delta > 0$ surrounding $\hat{F}$, i.e.

$$\mathcal{B}(\hat{F}, \delta) := \{G : G \text{ is a probability distribution on } \mathbb{R}^d \text{ s.t. } d_w(G, \hat{F}) \leq \delta\}$$

with d_w defined as

$$d_w(G_1, G_2) := \inf \left\{ E_P[\|\xi - \xi'\|] : \begin{array}{l} P \text{ is a joint probability distribution of } (\xi, \xi') \in \mathbb{R}^d \times \mathbb{R}^d \\ \text{with marginals } G_1 \text{ and } G_2 \text{ for } \xi \text{ and } \xi' \text{ respectively} \end{array} \right\}$$

where $\|\cdot\|$ is a norm on $\mathbb{R}^d$.

Exact Reformulation of (3): The distributionally robust chance constraint $Z := \{x \in \mathbb{R}^d : \inf_{G \in \mathcal{B}(\hat{F}, \delta)} P_G(\xi^T x \leq b) \geq 1 - \alpha\}$ in (3), if $b \geq 0$, has the following equivalent reformulation (Theorem 2 in Xie (2019))

$$Z = \left\{ x \in \mathbb{R}^d : \begin{array}{l} \delta v + \frac{1}{n} \sum_{i=1}^n z_i \leq \alpha r \\ r \leq z_i + s_i \text{ for } i = 1, \dots, n \\ s_i \leq b - \xi_i^T x + M(1 - y_i) \text{ for } i = 1, \dots, n \\ s_i \leq M y_i \text{ for } i = 1, \dots, n \\ \|x\|_* \leq v \\ v \geq 0, r \geq 0, s_i \geq 0, z_i \geq 0, y_i \in \{0, 1\} \text{ for } i = 1, \dots, n \end{array} \right\} \quad (4)$$

where $\|\cdot\|_*$ is the dual norm of $\|\cdot\|$, and M can be any number such that

$$M \geq \max_{i=1, \dots, n} \max_{x: 0 \leq x_l \leq 1 \text{ for each } l} |b - \xi_i^T x|.$$

For example, we can use $M = |b| + \max_{i=1, \dots, n} \|\xi_i\|_1$, with $\|\cdot\|_1$ denoting the l_1 -norm. Therefore (3) can be solved as a mixed integer program (Xie (2019) uses solvers from Gurobi).

Convex Approximation of (3): The exact constraint (4) is usually non-convex. A convex approximation of (4) based on CVaR is the following (Theorem 6 in Xie (2019))

$$Z_{CVaR} = \left\{ x \in \mathbb{R}^d : \begin{array}{l} \delta v + \frac{1}{n} \sum_{i=1}^n z_i \leq \alpha r \\ r \leq z_i + b - \xi_i^T x \text{ for } i = 1, \dots, n \\ \|x\|_* \leq v \\ v \geq 0, r \geq 0, z_i \geq 0 \text{ for } i = 1, \dots, n \end{array} \right\}. \quad (5)$$

It holds that $Z_{CVaR} \subseteq Z$, therefore a conservative convex approximation.

2.2 Problem (2)

The Wasserstein DRO formulation for (2) is

$$\begin{aligned}
& \min_x \quad c^T x \\
& \text{s.t.} \quad \sup_{Q \in \mathcal{B}(\hat{F}, \delta)} \inf_{t \in \mathbb{R}} t + \frac{1}{\alpha} E_Q[\max\{-\xi^T x - t, 0\}] \leq \gamma \\
& \quad \sum_{l=1}^d x_l = 1, x_l \geq 0 \text{ for each } l = 1, \dots, d
\end{aligned} \tag{6}$$

Exact Reformulation of (6): The robust CVaR constraint $Z := \{x \in \mathbb{R}^d : \sup_{Q \in \mathcal{B}(\hat{F}, \delta)} \inf_{t \in \mathbb{R}} t + \frac{1}{\alpha} E_Q[\max\{-\xi^T x - t, 0\}] \leq \gamma\}$, which can be reformulated as (see the CVaR approximation (5) of (3), or Corollary 5.1 in Esfahani and Kuhn (2018))

$$Z = \left\{ x \in \mathbb{R}^d : \begin{array}{l} \delta v + \frac{1}{n} \sum_{i=1}^n z_i \leq \alpha r \\ r \leq z_i + \gamma + \xi_i^T x \text{ for } i = 1, \dots, n \\ \|x\|_* \leq v \\ v \geq 0, r \geq 0, z_i \geq 0 \text{ for } i = 1, \dots, n \end{array} \right\}. \tag{7}$$

3 Validation Schemes

We consider three cross validation schemes to select the hyperparameter δ , the radius of the Wasserstein ball.

The three schemes are described in Algorithms 1-3, using the chance-constrained problem as an example. The CVaR-constrained problem can be validated in a similar way.

4 To-Do List

1. Implement a function that takes in the radius δ and other problem parameters such as $c, b, 1 - \alpha$ etc., and outputs the optimal solution of (3). The optimization is solved using either the mixed integer formulation (4) or the convex approximation (5).
2. Implement Algorithms 1-3 to select the radius δ .
3. Repeat 1. and 2. for the CVaR-constrained portfolio problem (6), using the robust reformulation (7).

References

- Esfahani PM, Kuhn D (2018) Data-driven distributionally robust optimization using the wasserstein metric: Performance guarantees and tractable reformulations. *Mathematical Programming* 171(1-2):115–166.
- Xie W (2019) On distributionally robust chance constrained programs with wasserstein distance. *Mathematical Programming* 1–41.

Algorithm 1 K-Fold Cross Validation

Input: n observations $\Xi = \{\xi_1, \dots, \xi_n\}$; a set of candidate parameter values $\{\delta_1, \dots, \delta_m\}$; a confidence level $1 - \beta$; number of folds K .

Procedure:

1. Randomly partition the data into K subsets of equal size, $\Xi_k, k = 1, \dots, K$ such that each Ξ_k contains n/K data and $\Xi = \cup_{k=1}^K \Xi_k$.
2. For each $k = 1, \dots, K$ and each $j = 1, \dots, m$, use all the data except the k -th group to solve (3) with $\delta = \delta_j$ and $\hat{F} = \frac{K}{(K-1)n} \sum_{\xi \in \Xi \setminus \Xi_k} \delta_\xi$ to obtain its optimal solution $x_k^*(\delta_j)$, and then estimate the chance constraint $\hat{P}_k(\delta_j) = \frac{K}{n} \sum_{\xi \in \Xi_k} \mathbf{1}\{\xi^T x_k^*(\delta_j) \leq b\}$.
3. Compute

$$\delta^* = \underset{\delta_j, 1 \leq j \leq m}{\operatorname{argmin}} \left\{ \frac{1}{K} \sum_{k=1}^K c^T x_k^*(\delta_j) : \frac{\sum_{k=1}^K \mathbf{1}\{\hat{P}_k(\delta_j) \geq 1 - \alpha\}}{K} \geq 1 - \beta, j = 1, \dots, m \right\}$$

4. Use all the data Ξ to resolve (3) with $\delta = \delta^*$ and $\hat{F} = \frac{1}{n} \sum_{i=1}^n \delta_{\xi_i}$ to obtain its optimal solution $x^*(\delta^*)$

Output: The solution $x^*(\delta^*)$.

Algorithm 2 Bootstrapping (Section 7.2.3 in Esfahani and Kuhn (2018))

Input: n observations $\Xi = \{\xi_1, \dots, \xi_n\}$; a set of candidate parameter values $\{\delta_1, \dots, \delta_m\}$; a confidence level $1 - \beta$; number of bootstrap resamples B .

Procedure:

1. For each $b = 1, \dots, B$, uniformly draw n data with replacement from the data set Ξ to get the resampled data $\xi_1^b, \dots, \xi_n^b$, and let $\Xi_b^c = \Xi \setminus \{\xi_1^b, \dots, \xi_n^b\}$ be the remaining data that are not resampled.
2. For each $b = 1, \dots, B$ and each $j = 1, \dots, m$, use the b -th resample $\xi_1^b, \dots, \xi_n^b$ to solve (3) with $\delta = \delta_j$ and $\hat{F} = \frac{1}{n} \sum_{i=1}^n \delta_{\xi_i^b}$ to obtain its optimal solution $x_b^*(\delta_j)$, and then estimate the chance constraint $\hat{P}_b(\delta_j) = \frac{1}{\operatorname{card}(\Xi_b^c)} \sum_{\xi \in \Xi_b^c} \mathbf{1}\{\xi^T x_b^*(\delta_j) \leq b\}$, where $\operatorname{card}(\cdot)$ denotes the cardinality of a set.
3. Compute

$$\delta^* = \underset{\delta_j, 1 \leq j \leq m}{\operatorname{argmin}} \left\{ \frac{1}{B} \sum_{b=1}^B c^T x_b^*(\delta_j) : \frac{\sum_{b=1}^B \mathbf{1}\{\hat{P}_b(\delta_j) \geq 1 - \alpha\}}{B} \geq 1 - \beta, j = 1, \dots, m \right\}$$

4. Use all the data Ξ to resolve (3) with $\delta = \delta^*$ and $\hat{F} = \frac{1}{n} \sum_{i=1}^n \delta_{\xi_i}$ to obtain its optimal solution $x^*(\delta^*)$

Output: The solution $x^*(\delta^*)$.

Algorithm 3 Sectioning (Section 5.2 in Xie (2019))

Input: n observations $\Xi = \{\xi_1, \dots, \xi_n\}$; training size n_1 and validation size n_2 ($n_1 + n_2 = n$); number of sections K ; a set of candidate parameter values $\{\delta_1, \dots, \delta_m\}$; a confidence level $1 - \beta$.

Procedure:

1. Randomly split Ξ into Ξ_t and Ξ_v such that $\text{card}(\Xi_t) = n_1$ and $\text{card}(\Xi_v) = n_2$, and further split Ξ_v evenly into $\Xi_{v,k}, k = 1, \dots, K$ such that $\text{card}(\Xi_{v,k}) = n_2/K$ for each k .
2. For each $j = 1, \dots, m$, use the training set Ξ_t to solve (3) with $\delta = \delta_j$ and $\hat{F} = \frac{1}{n_1} \sum_{\xi \in \Xi_t} \delta_\xi$ to obtain its optimal solution $x^*(\delta_j)$, and then for each $k = 1, \dots, K$ estimate the chance constraint $\hat{P}_k(\delta_j) = \frac{1}{\text{card}(\Xi_{v,k})} \sum_{\xi \in \Xi_{v,k}} \mathbf{1}\{\xi^T x^*(\delta_j) \leq b\}$.
3. Compute

$$\delta^* = \underset{\delta_j, 1 \leq j \leq m}{\text{argmin}} \left\{ c^T x^*(\delta_j) : \frac{\sum_{k=1}^K \mathbf{1}\{\hat{P}_k(\delta_j) \geq 1 - \alpha\}}{K} \geq 1 - \beta, j = 1, \dots, m \right\}$$

Output: The solution $x^*(\delta^*)$.
